

# Task 4: AI Hand Gesture Computer Control System

|                |                                              |
|----------------|----------------------------------------------|
| Submitted By:  | K. Ameesha                                   |
| Task ID:       | Task 4 - Computer Vision & OS Automation     |
| Tech Stack:    | Python, OpenCV, MediaPipe, PyAutoGUI         |
| Target Domain: | Real-Time Touchless Operating System Control |

## 1. Introduction & Project Overview

Building upon standard static computer vision classification, this project implements an active **AI Hand Gesture Computer Control System**. By utilizing a standard webcam input, the system continuously tracks human hand landmarks and translates precise spatial gestures into operating system-level commands, allowing users to control cursor movement, clicking, and scrolling without physical contact with peripherals.

## 2. Project Objectives

The primary objective of this task is to construct a robust, real-time automation pipeline that bridges computer vision hand tracking with OS-level event execution:

- **Real-Time Cursor Navigation:** Map index finger landmark coordinates directly to monitor screen resolution for smooth pointer movement.
- **Touchless Clicking Mechanism:** Detect finger distance thresholds (e.g., thumb-to-index pinch) to trigger mouse click operations programmatically.
- **System Control Integration:** Maintain low latency and high accuracy across frame updates to ensure practical usability.

## 3. Technology Stack & Libraries

- **Python:** Core programming language utilized for application orchestration and script execution.
- **OpenCV (cv2):** Handles live webcam feed capture, image processing transformations, and real-time GUI frame rendering.
- **MediaPipe Hands:** Google's cross-platform machine learning framework optimized for high-fidelity 3D hand landmark coordinate estimation.

- **PyAutoGUI:** Cross-platform GUI automation library used to programmatically control mouse cursor positioning, clicking, and system actions.

## 4. Pipeline & Implementation Methodology

The execution workflow is structured into continuous real-time iterative blocks:

- **Frame Capture & Mirroring:** Reads sequential frames from `cv2.VideoCapture(0)` and flips them horizontally for intuitive mirror-image visualization.
- **Color Space Conversion:** Transforms input BGR video frames into RGB color format as required by the MediaPipe Hands processing pipeline.
- **Landmark Extraction:** Detects 21 3D hand joint coordinates on processed frames to identify key tracking points (such as Index Finger Tip and Thumb Tip).

## 5. Feature Extraction & Control Logic

The system evaluates coordinate thresholds and geometric distance formulas across landmark joints to execute system controls:

| Gesture / Action | Target Landmarks                  | Heuristic Rule / Condition                                       | Behavioral Description                                                       |
|------------------|-----------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------|
| Cursor Movement  | Index Finger Tip (ID 8)           | Normalized X & Y coordinates scaled to screen width and height.  | Moves the mouse pointer smoothly across the monitor following finger motion. |
| Click Trigger    | Index Tip & Thumb Tip (IDs 8 & 4) | Euclidean distance $d < 35$ pixels between thumb and index tips. | Executes a programmatic mouse click event with built-in debounce delay.      |

## 6. Step-by-Step Execution Guide

To run the application locally in your environment:

```
# 1. Install required dependencies
pip install opencv-python mediapipe pyautogui

# 2. Place script file (controller.py) inside your task folder
# 3. Execute application via terminal
python controller.py
```

## 7. Results & Visual Verification

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The application successfully renders real-time skeleton annotations on the live video feed while simultaneously controlling OS cursor position and click events based on physical hand motions.

**Status: Completed & Verified**